

NORTHWIND SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI

Project Report

Submitted By

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Tool Used: Microsoft Power BI Desktop

1. Introduction

For this project, I created an interactive sales dashboard in Microsoft Power BI using the Northwind dataset. The dashboard combines order, customer, employee, product and category information to present sales activity through KPI cards, charts, a detailed product table and interactive filters. The aim is to convert the underlying business data into a clear visual report that can be explored quickly.

The report includes an overall sales overview, product and customer analysis, and an interactive sales dashboard. The visuals update according to selections made through the available year, region and category filters.

2. Objectives

The main objectives of the dashboard were to:

- Summarise total orders, freight, customers and quantity sold using KPI cards.
- Analyse order activity across months.
- Compare sales performance across product categories.
- Analyse orders handled by employees.
- Study customer distribution across regions.
- Provide an interactive dashboard using filters and slicers.

3. Dataset Description

The dashboard uses the Northwind business dataset. The Power BI data model shown in the screenshots contains the following tables:

- Categories — product category information.
- Customers — customer and regional information.
- Employees — employee information used for order analysis.
- Order details — detailed order-line information.
- Orders — order-level information.
- Products — product names and category-related information.
- Shippers — shipping-related information.

4. Data Preparation

The dashboard was prepared by loading the Northwind tables into Power BI Desktop and using the available fields to build the required visuals. The report uses fields from orders, order details, customers, employees, products and categories so that the dashboard can present both summary and detailed views.

- Imported the Northwind data into Power BI Desktop.
- Used order and order-detail information for order and quantity analysis.
- Used customer information for regional customer analysis.
- Used employee information to compare order activity by employee.
- Used product and category information for product-level and category-level analysis.
- Configured report visuals and filters so that selections can update the dashboard.

5. Measures / KPI Values Used

The dashboard presents the following key measures as KPI cards. The exact displayed values change when interactive filters are applied:

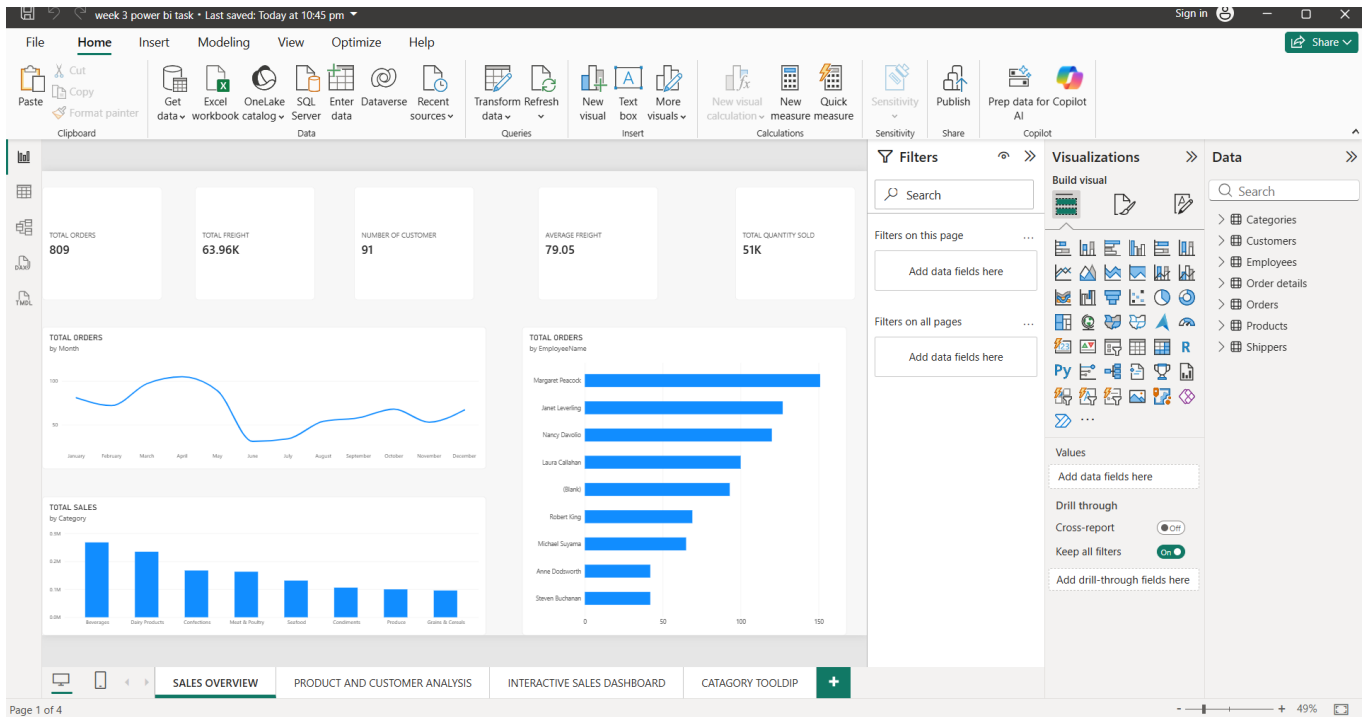
- Total Orders
- Total Freight
- Number of Customer
- Average Freight
- Total Quantity Sold

6. Dashboard Components

The completed Power BI report contains four main dashboard views. Each view provides a different way of analysing the Northwind data.

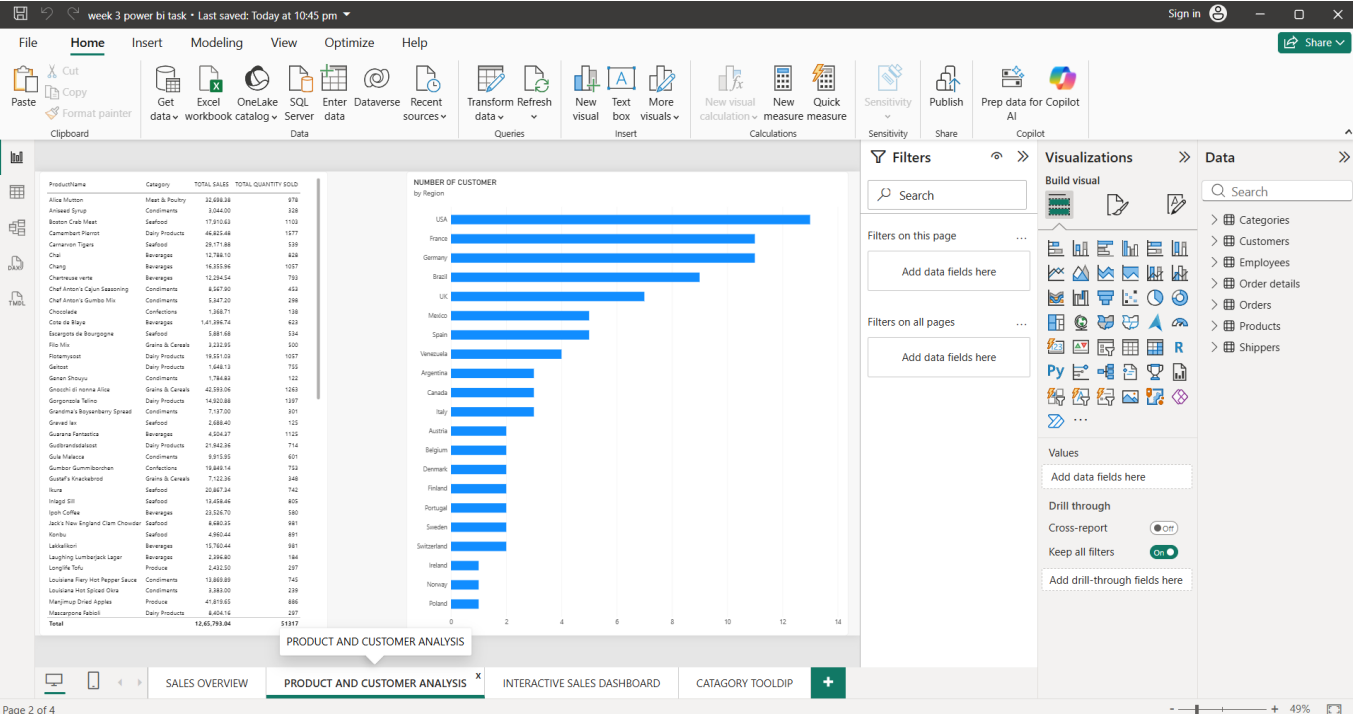
- Sales Overview — KPI cards, monthly order trend, employee-wise orders and category-wise sales.
- Product and Customer Analysis — product-level table and customer count by region.
- Interactive Sales Dashboard — year, region and category filters together with KPI cards and charts.
- Interactive Dashboard View — an additional captured view of the same interactive dashboard layout.

Sales Overview



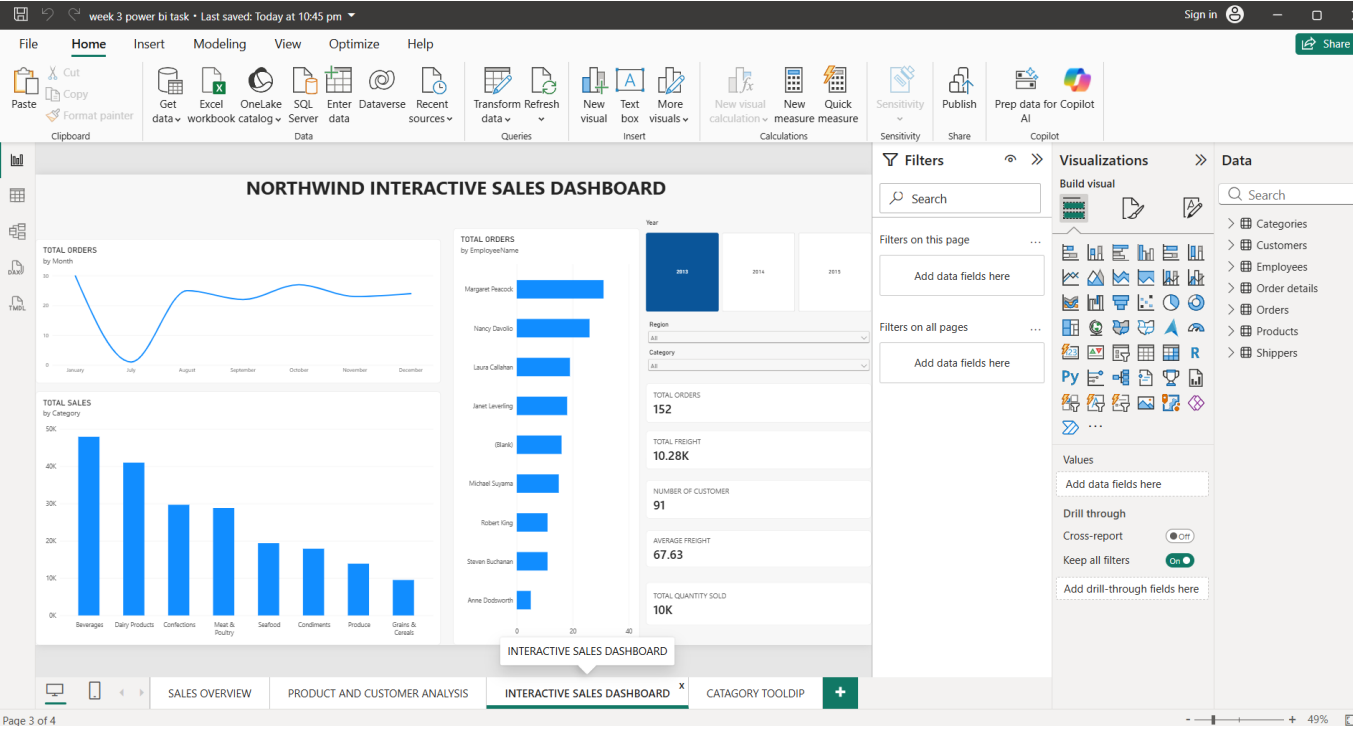
The Sales Overview page gives a high-level summary of the Northwind data. The screenshot shows 809 total orders, 63.96K total freight, 91 customers, 79.05 average freight and 51K total quantity sold. It also contains a monthly order trend, employee-wise order comparison and category-wise total sales.

Product and Customer Analysis



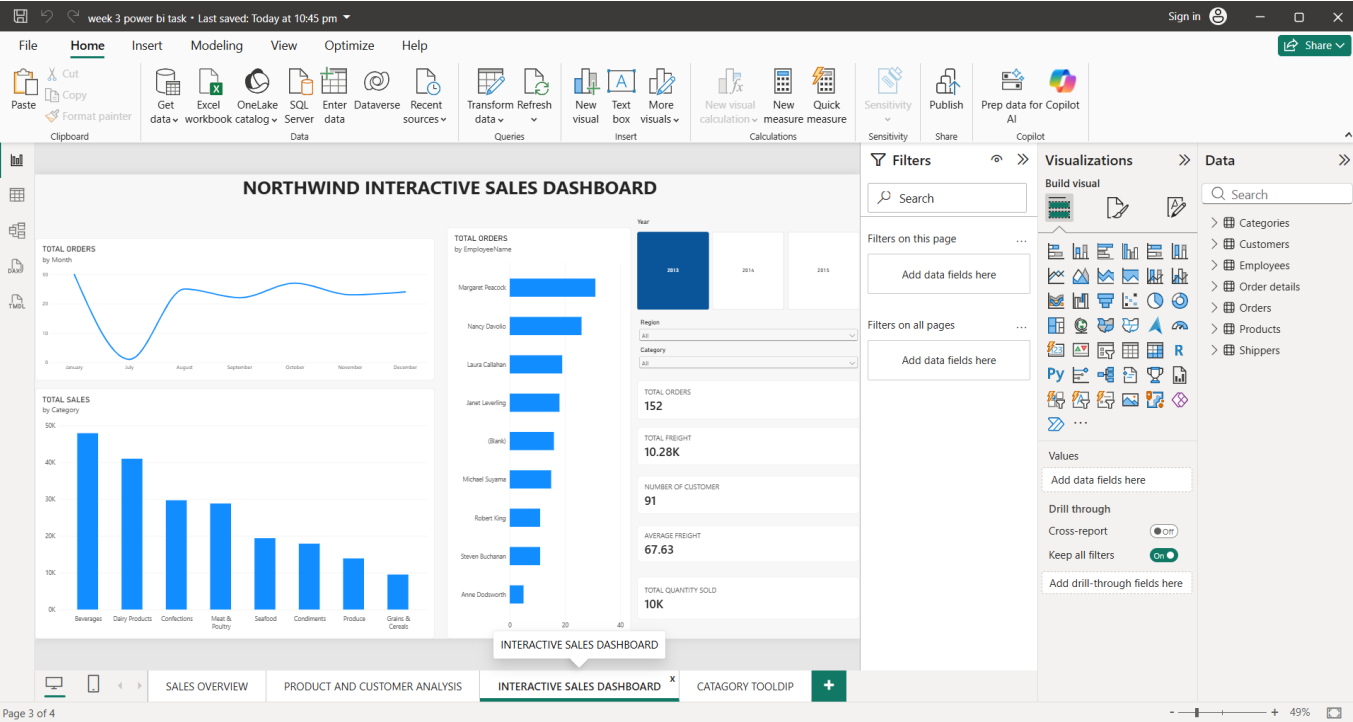
This page combines a detailed product table with customer analysis by region. The table displays Product Name, Category, Total Sales and Total Quantity Sold, while the regional bar chart shows the number of customers across different countries.

Interactive Sales Dashboard



The interactive dashboard combines a monthly order trend, category-wise sales, employee-wise orders, year selection and Region and Category filters. With 2013 selected in the screenshot, the KPI cards show 152 total orders, 10.28K total freight, 91 customers, 67.63 average freight and 10K total quantity sold.

Interactive Sales Dashboard – Captured View



This additional screenshot captures the interactive dashboard layout and its main controls. It shows the same report structure with the year slicer, Region and Category filters, employee order chart, monthly order trend, category sales chart and KPI cards.

7. Slicers and Interactive Filters

The interactive dashboard uses filters to allow the user to focus on a selected portion of the Northwind data. The controls visible in the screenshots include:

- Year — used to select the required year, with 2013 selected in the captured view.
- Region — used to filter the dashboard according to region.
- Category — used to filter the dashboard according to product category.

These controls make the dashboard interactive because the displayed charts and KPI values can be examined under different selections instead of remaining as one fixed summary.

8. Key Findings

The screenshots provide several direct observations from the completed dashboard:

- The overall Sales Overview displays 809 total orders.
- Total freight in the overall view is displayed as 63.96K.
- The dashboard displays 91 customers in the overall view.
- Average freight in the overall view is 79.05.
- Total quantity sold is displayed as 51K in the overall view.
- The monthly order chart shows that order activity changes across the months.
- The employee chart shows differences in order activity among employees.
- The category chart shows different sales contributions from the available product categories.
- The regional customer chart shows that customer counts vary considerably across countries.
- When 2013 is selected in the interactive dashboard, the displayed total orders are 152 and total freight is 10.28K.

These observations demonstrate how the same underlying dataset can be viewed from overall, product, customer, employee, category and time-based perspectives.

9. Conclusion

The Northwind Sales Performance Dashboard demonstrates how Microsoft Power BI can transform business data into an interactive visual report. The combination of KPI cards, line charts, bar charts, a detailed product table and interactive filters makes the information easier to explore than a plain data table.

The project also provides practical experience in organising data, selecting suitable visualisations and using filters to examine information from different perspectives. The final dashboard brings together sales, order, customer, employee, product and category analysis in one report.

Overall, the dashboard provides a clear visual summary of the Northwind data and demonstrates the use of Power BI for interactive business analysis.

10. References

- Microsoft Power BI Desktop
- Northwind Dataset
- Microsoft Power BI documentation

End of Report